Gastrointestinal Physiology Oral Cavity & Salivation
GIT L1

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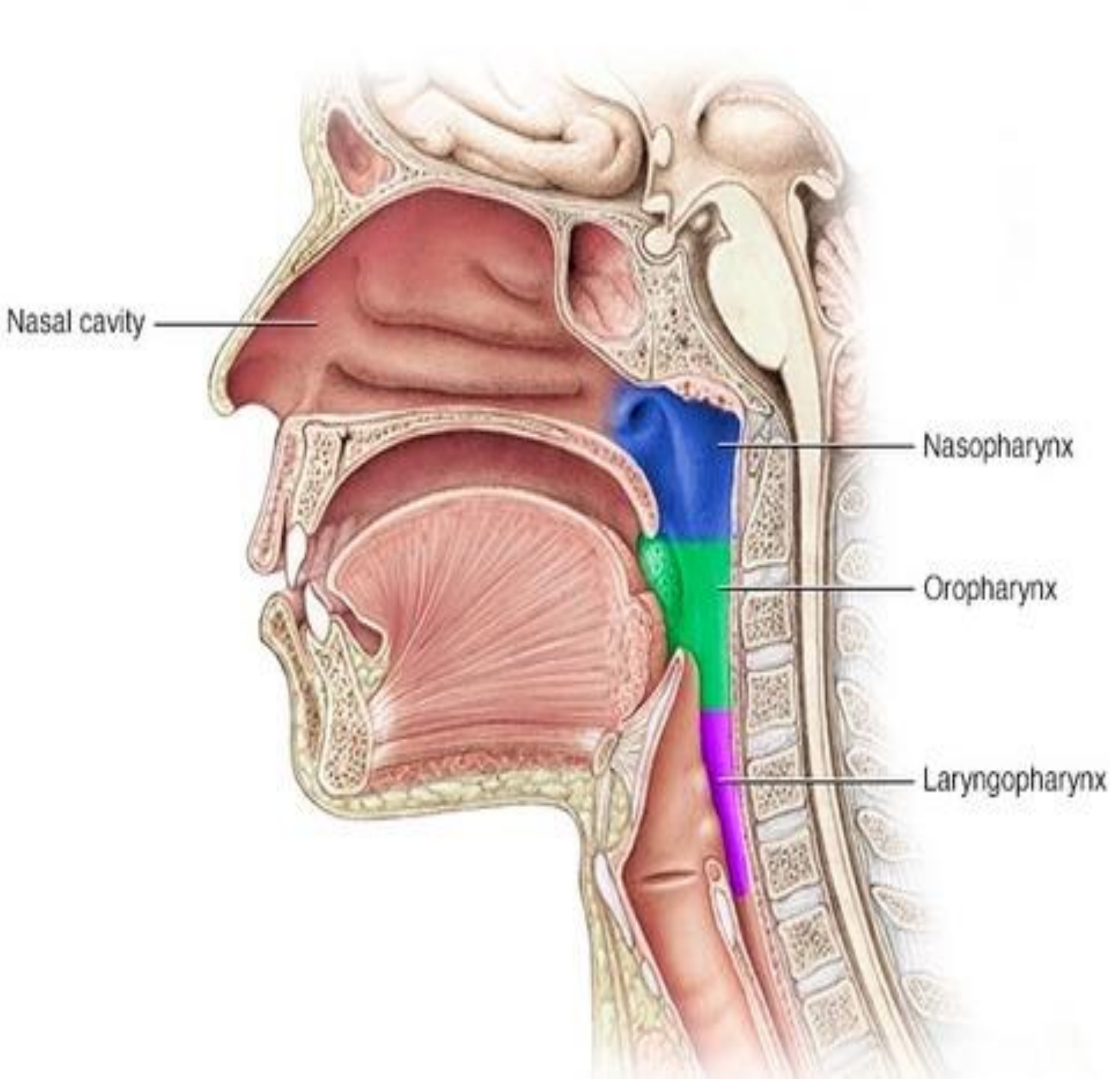
Learning Outcomes

1. Describe the structural organization and main functions of the **oral cavity**.
2. Explain the role of **mastication and taste** in the initiation of digestion.
3. Discuss the **major salivary glands** and their contributions to saliva formation.
4. Explain the **composition and functions of saliva** in digestion, lubrication, and defense.
5. Understand the **neural control of salivary secretion** and its clinical relevance.

Oral Cavity Physiology

Structure & Functional Overview

- The oral cavity is the **first segment of the gastrointestinal tract (GIT)**. It provides both **mechanical and chemical digestion**, as well as a key role in **sensory perception and defense**.



Functions of the Oral Cavity

1. Mastication (Chewing):

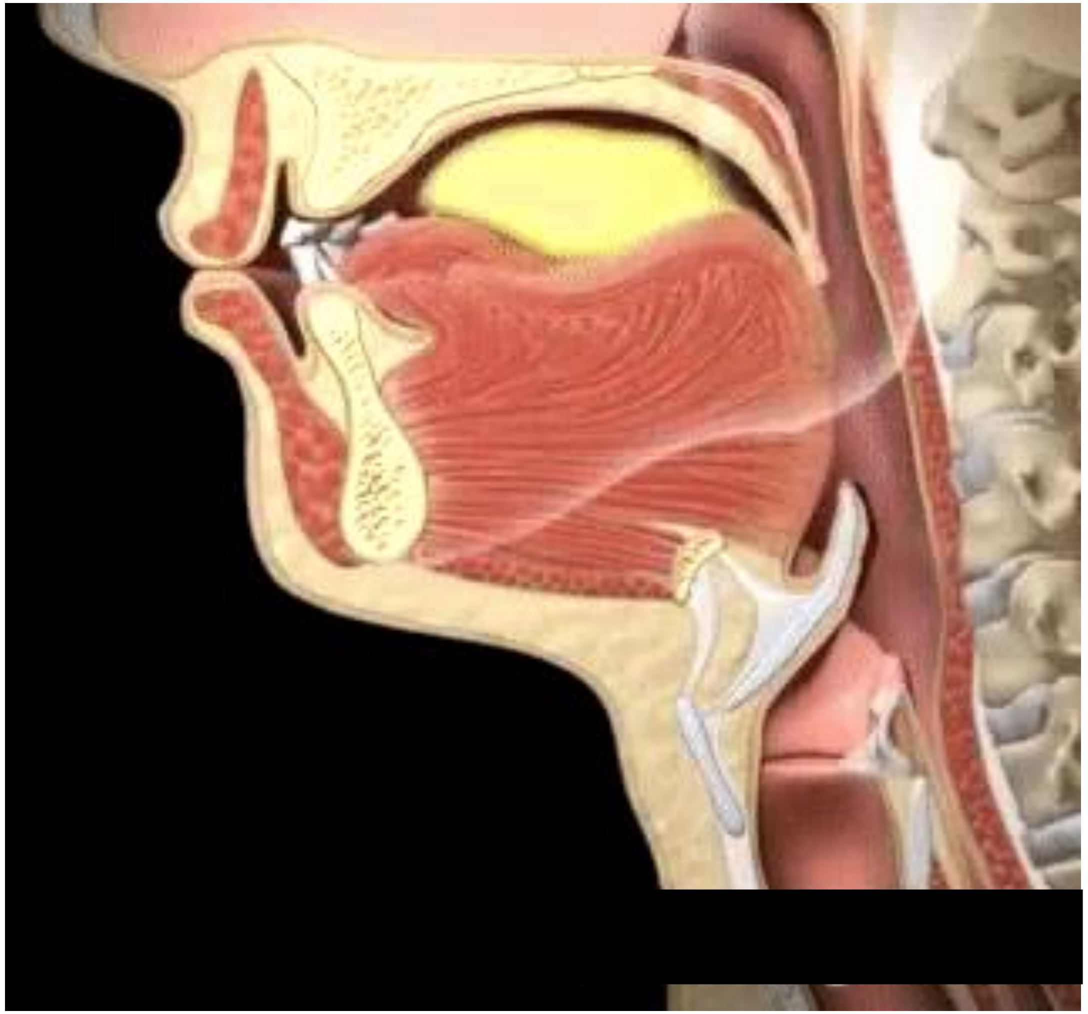
1. Increases food surface area for enzymatic digestion.
2. Controlled by reflexes (periodontal pressure receptors → brainstem chewing center → motor nuclei of V, VII, IX, XII).
3. Muscles: masseter, temporalis, medial & lateral pterygoids.

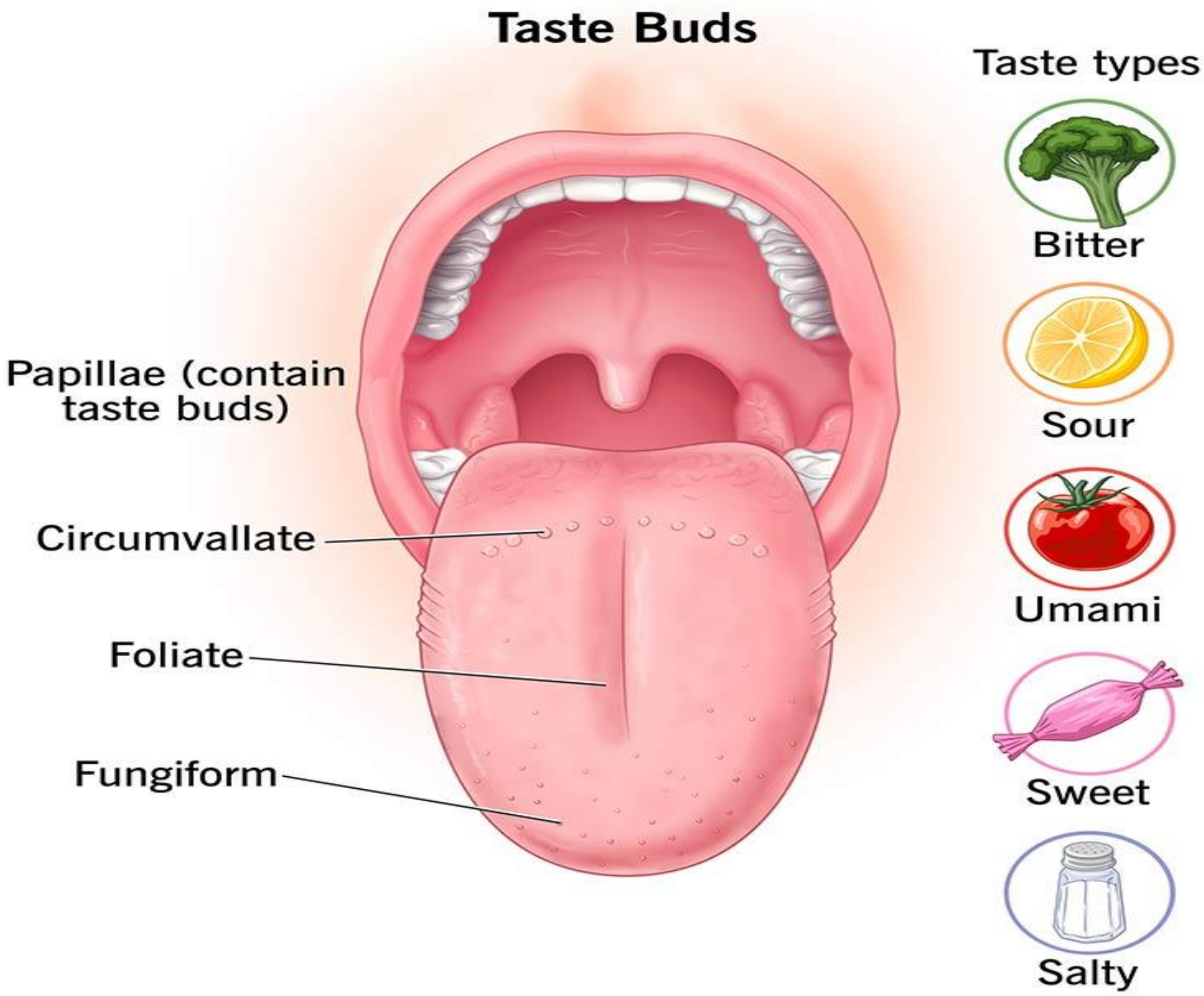
2. Taste & Sensory Analysis:

1. Taste buds located mainly on tongue papillae (fungiform, circumvallate, foliate).
2. Modalities: sweet, sour, salty, bitter, umami.
3. Neural pathways:
   1. CN VII (anterior 2/3),
   2. CN IX (posterior 1/3),
   3. CN X (epiglottis, pharynx).
4. Taste signals → nucleus tractus solitarius → hypothalamus & salivatory nuclei → stimulate salivation.

Mastication / Muscle Action

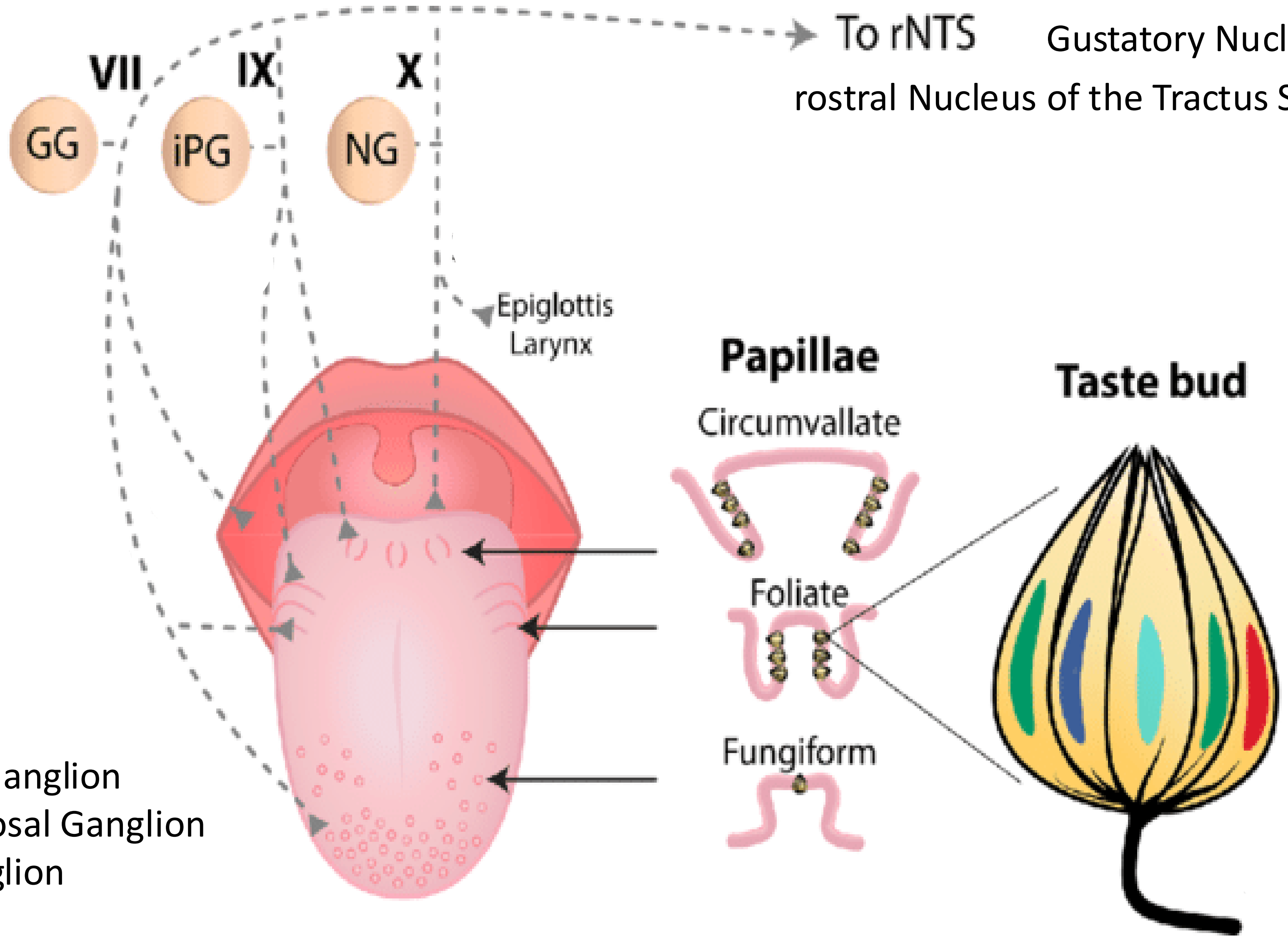
Chewing / Mastication




Taste Buds
Taste types
Bitter
Sour
Umami
Sweet
Salty
Papillae (contain taste buds)
Circumvallate
Foliate
Fungiform

Gustatory Nucleus

rostral Nucleus of the Tractus Solitarius



GG – Geniculate Ganglion
iPG - Inferior Petrosal Ganglion
NG - Nodose Ganglion

3. Lubrication & Bolus Formation:

- Saliva moistens food.
- Mucins coat particles, facilitating bolus formation and safe swallowing.

4. Defense:

- Tonsillar tissue in provides immune surveillance.
- Salivary IgA, lysozyme, lactoferrin protect against pathogens.

5. Speech:

- Tongue and lips manipulate airflow and resonance for articulation.

Salivation

Anatomy of Salivary Glands

Major glands:

Parotid (serous, watery, rich in amylase)., Submandibular (mixed, 70% of saliva)., Sublingual (mucous-rich).

Minor glands: dispersed in lips, cheeks, palate, tongue.

Composition of Saliva

Water (~99%).

Electrolytes: Na^+, K^+, Cl^-, HCO_3^-, Ca^{2+}, phosphate.

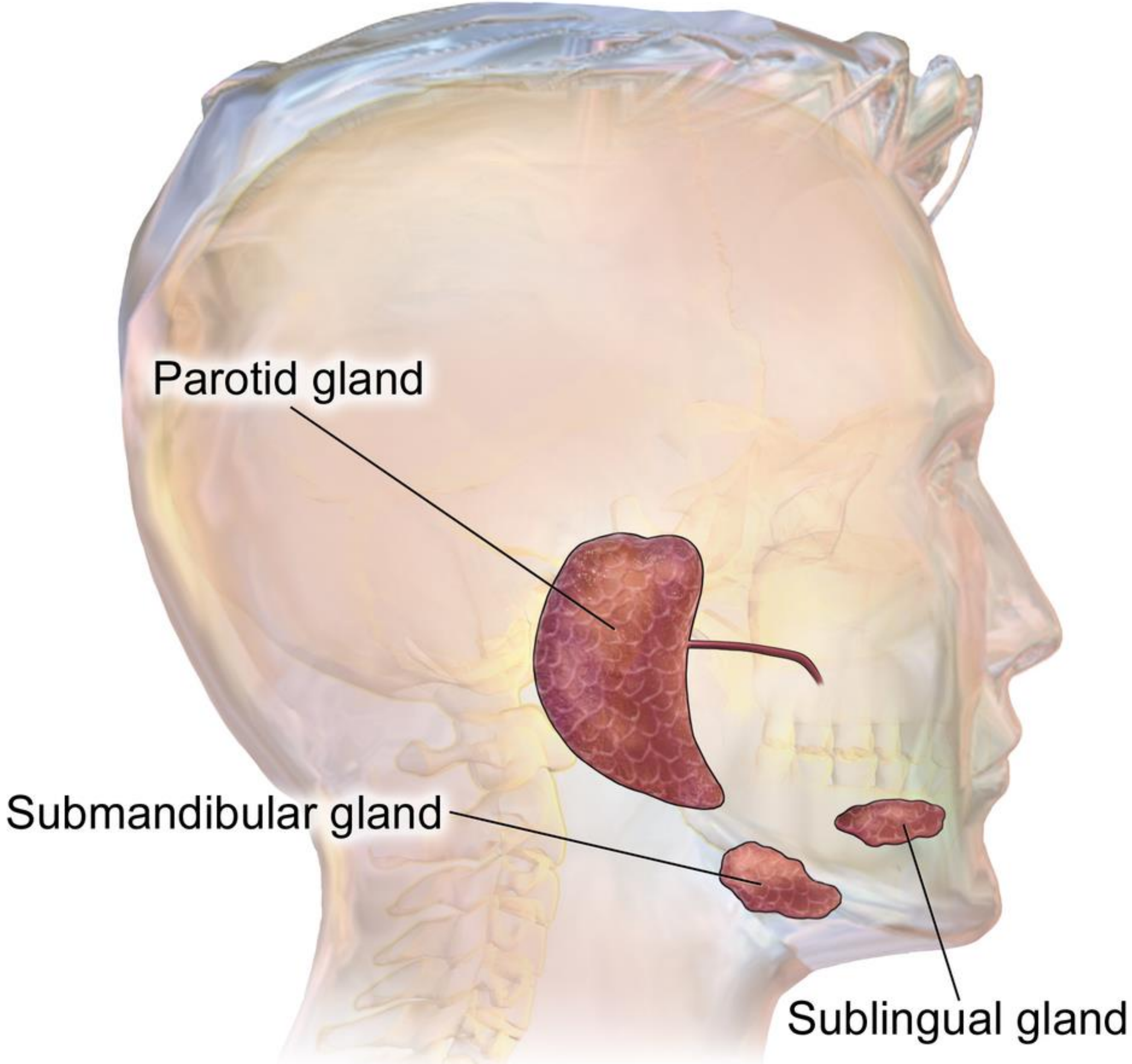
Proteins/Enzymes:

α-Amylase (starch digestion). , Lingual lipase (fat digestion, active in stomach). , Kallikrein (forms bradykinin → increases blood flow).

Mucins: lubrication.

Antimicrobial factors: lysozyme, lactoferrin, peroxidase, IgA.

Excretory products: urea, uric acid, some drugs.



Salivary Glands

Physiology of Saliva Formation

- **Primary secretion (acinus):** isotonic, plasma-like fluid containing amylase and mucin.
- **Ductal modification:**
  - Na^+ and Cl^- actively reabsorbed.
  - K^+ and HCO_3^- secreted.
  - Ducts impermeable to water → final saliva becomes **hypotonic**.
- Flow rate affects composition:
  - **Low flow:** more hypotonic, acidic (↑K^+).
  - **High flow:** closer to plasma composition, alkaline (↑Na^+, HCO_3^-).

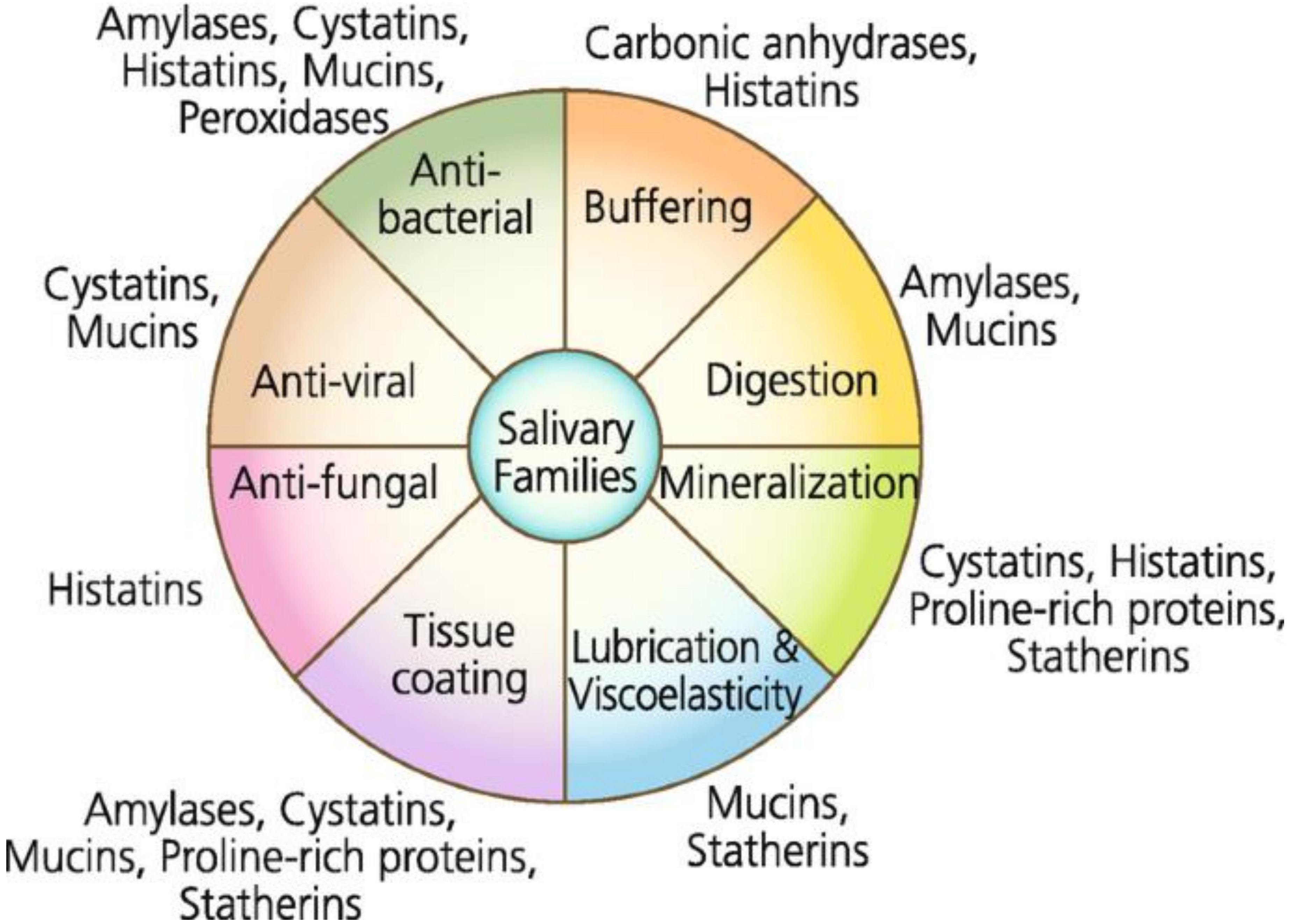
Control of Salivary Secretion

Neural control dominates; no hormonal regulation.

- **Parasympathetic:**
  - CN VII (chorda tympani → submandibular ganglion).
  - CN IX (otic ganglion → parotid).
  - Ach → muscarinic M3 receptors → copious watery saliva.
- **Sympathetic:**
  - T1–T3 → superior cervical ganglion → NE → β-adrenergic receptors.
  - Secretion is smaller in volume, more mucous and protein-rich.

Functions of Saliva

1. **Digestion:** begins carbohydrate & lipid digestion.
2. **Lubrication:** for swallowing & speech.
3. **Protection:** antimicrobial, buffer, remineralization of teeth.
4. **Taste:** dissolves substances for chemoreceptors.
5. **Excretion:** minor route for some waste.


Amylases, Cystatins, Histatins, Mucins, Peroxidases
Carbonic anhydrases, Histatins
Anti-bacterial
Buffering
Cystatins, Mucins
Amylases, Mucins
Anti-viral
Digestion
Salivary Families
Anti-fungal
Mineralization
Histatins
Cystatins, Histatins, Proline-rich proteins, Statherins
Tissue coating
Lubrication & Viscoelasticity
Amylases, Cystatins, Mucins, Proline-rich proteins, Statherins
Mucins, Statherins